Garage Door Automation Project
Programmable Logic Controller

Project

In this project, you will be provided with a downsized model of a garage door setup. The garage door has a 12V DC motor, a 12V light bulb, and two sensors each comprising a 5V proximity sensor and a 12V relay. In this project you will:

1) Do all the wiring needed to connect the downsized garage door to Click PLC in order to control such door according to the specifications below. You will use the relay module C0 04TRS, the input module C0 08ND3 and an output module, either the C0 08TD1 or C0 08TD1.

2) Program the PLC to control the garage door with the following specifications:

1. The green button connected to the PLC will be the operation button that will control the motor motion. With the green button alone, you should move the door up, down and stop it.
2. There should be a power-up toggle switch. When the switch is on, the door and all its elements operate normally. When the switch is off, the door and all its elements are off and do not operate.
3. At power-up, the door will close if it's partially or fully open.
4. If the door is stopped and the operation button is pressed, the door will move in the direction opposite to the last motion. If the door is moving and the operation button is pressed, the door will stop. (For example: the door is fully open (which means the previous motion was up), then, if the operation button is pressed, the door will go down to close. During the going down motion, if the operation button is pressed again, the door will stop. If the button is pressed again, the door will go up).
5. While the door is in motion, the garage door light (12V light bulb) is on.
6. When the door stops, the garage light stays on for 15 seconds.

3) Write a very formal report:

1. State the problem with its specifications using your own words. Describe your hardware: inputs, outputs, PLC used. With this information, your reader will know what you were up to.
2. Describe your wiring and present schematics of your wiring. Describe the sensors you are using and show their schematics and how they work.

3. Stage 1: DESIGN. Design your implementation with an FSM bubble diagram with states and transitions. Follow the “Rules of Engagement” which were described in class and are written in the slides to draw and describe FSMs.
4. Stage 2: IMPLEMENTATION. Follow the structured method taught in class. The grading of the implementation is based on the adherence of the implementation rules explained in class and the specification compliance. Remember to label all components and comment all your rungs of your ladder diagram.
5. Stage 3: TEST. You will establish a set of tests (test suite) and document their results with respect to the expected results. Verify that all specifications are satisfied. Use a table for your test suite.
6. Conclusions and Recommendations.
7. Final Note: Give your report a professional look. (Handwriting drawings and stuff do not have a professional look).